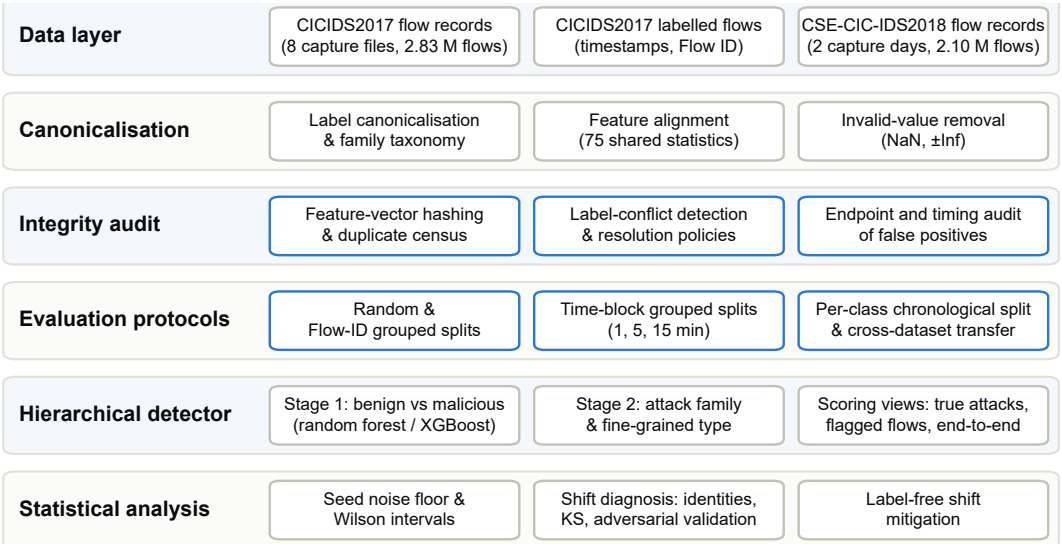
Data layer
CICIDS2017 flow records (8 capture files, 2.83 M flows)
CICIDS2017 labelled flows (timestamps, Flow ID)
CSE-CIC-IDS2018 flow records (2 capture days, 2.10 M flows)
Canonicalisation
Label canonicalisation & family taxonomy
Feature alignment (75 shared statistics)
Invalid-value removal (NaN, ±Inf)
Integrity audit
Feature-vector hashing & duplicate census
Label-conflict detection & resolution policies
Endpoint and timing audit of false positives
Evaluation protocols
Random & Flow-ID grouped splits
Time-block grouped splits (1, 5, 15 min)
Per-class chronological split & cross-dataset transfer
Hierarchical detector
Stage 1: benign vs malicious (random forest / XGBoost)
Stage 2: attack family & fine-grained type
Scoring views: true attacks, flagged flows, end-to-end
Statistical analysis
Seed noise floor & Wilson intervals
Shift diagnosis: identities, KS, adversarial validation
Label-free shift mitigation